

Topic: Spatio-Temporal Variability of the Water Cycle

Goal: Gain familiarity with analysis tools for estimating long-term trends and seasonal variability in NASA's multi-decade terrestrial hydrology products

Introduction: Continuous monitoring of the water cycle is critical for understanding many interconnected processes within the climate system. In recent decades, remote sensing has emerged as a critical observing technology to complement in situ observations of the terrestrial water cycle. NASA's Gravity Recovery and Climate Experiment (GRACE) and follow-on (GRACE-FO) missions provide an extended global data record of water cycle perturbations at regional and seasonal scales. This project will introduce tools for decomposing trends and seasonal cycles in water cycle dynamics using the missions' combined data record.

Datasets:

- GRACE/GRACE-FO global total water storage anomalies (TWSA)
 - Location:
`/home/jovyan/shared/NASA_Summer_School_2026/water_cycle/materials/data/GRCTellus.JPL.200204_202603.GLO.RL06.3M.MSCNV04CRI.nc`
- Example TWSA time series over central California
 - Location:
`/home/jovyan/shared/NASA_Summer_School_2026/water_cycle/materials/data/GRACEFO_TimeSeries_CA_2002_2026.csv`

Demo Scripts:

- `GRACE_NB1_GPTrend_Example.ipynb`: Estimating trend and seasonal cycle for GRACE/GRACE-FO time series
- `GRACE_NB2_GPFit_torch.ipynb`: Fit Gaussian process (GP) models to residuals
- `GRACE_NB3_GPPredict_torch.ipynb`: Explore correlation structure for GP fit and make predictions for chosen time points
- `GRACE_NB4_Bootstrap_torch.ipynb`: Example bootstrap procedure for trend and seasonal amplitude

Supporting Scripts:

- `GRACE_Map_TWS.ipynb`: Extract and map TWSA for selected region
- `GRACE_Extract_TS.ipynb`: Extract and save a TWSA time series for selected location

Questions:

- Regional analysis
 - Is a common seasonal cycle/trend sufficient for nearby locations within a region?
 - How can spatial correlation of residuals be evaluated?
- Contrast across regions/continents
 - Are temporal correlation length scales different across regions?

- What does that suggest about hydrologic processes in these areas?
- Which areas have the most substantial long-term trends?

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Project Server: <https://hub.jpl-cmda.org>